Unit 1

Introduction to Software Engineering

SOFTWARE PRODUCT AND PROCESS CHARACTERISTICS:

- **Software: -**
- Software is nothing but collection of computer programs and related documents that are planned to provide desired features, functionalities and better performance.

- Software is more than just a program code
- Software, when made for a specific requirement is called **software product.**

SOFTWARE PRODUCT AND PROCESS CHARACTERISTICS:

- **Engineering** on the other hand, is all about developing products, using well-defined, scientific principles and methods.

- **Software Engineering:** The application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software; that is, the application of engineering to software.

SOFTWARE PRODUCT AND PROCESS CHARACTERISTICS:

- Software product classified in 2 classes:

- 1. **Generic software:** developed to solution whose requirements are very common fairly stable and well understood by software engineer.
- 2. **Custom software:** developed for a single customer according to their specification.

Need of Software Engineering

- The need of software engineering arises because of higher rate of change in user requirements and environment on which the software is working.
- **Large software** - It is easier to build a wall than to a house or building, likewise, as the size of software become large engineering has to step to give it a scientific process.
- **Scalability**- If the software process were not based on scientific and engineering concepts, it would be easier to re-create new software than to scale an existing one.

Need of Software Engineering

- **Cost-** As hardware industry has shown its skills and huge manufacturing has lower down the price of computer and electronic hardware. But the cost of software remains high if proper process is not adapted.
- **Dynamic Nature-** The always growing and adapting nature of software hugely depends upon the environment in which user works. If the nature of software is always changing, new enhancements need to be done in the existing one. This is where software engineering plays a good role.
- **Quality Management-** Better process of software development provides better and quality software product.

Software Engineering Goals

- Readability (understood by those who maintain it)
- Correctness
- Reliability (high performance)
- Re usability
- Extensibility (ability to perform its operations)
- Flexibility
- Efficiency

Software Crisis

- Many software projects failed.
- Many software projects late, over budget, providing unreliable software that is expensive to maintain.
- Many software projects produced software which did not satisfy the requirements of the customer.
- Complexities of software projects increased as hardware Capability increased.
- Larger software system is more difficult and expensive to maintain.
- Demand of new software increased faster than ability to generate new software.

Characteristics of good software: -

- A software product can be judged by what it offers and how well it can be used. This software must satisfy on the following grounds:
- Operationa
- Transitional
- Maintenance

Operational: -

- This tells us how well software works in operations. It can be measured on:
- · Budget
- · Usability
- · Efficiency
- · Correctness
- · Functionality
- · Dependability
- · Security
- · Safety

Transitional: -

- This aspect is important when the software is moved from one platform to another:
- · Portability
- · Interoperability
- · Reusability
- · Adaptability

Maintenance: -

- This aspect briefs about how well software has the capabilities to maintain itself in the ever-changing environment:
- · Modularity
- · Maintainability
- · Flexibility
- · Scalability

Definition

- In short, Software engineering is a branch of computer science, which uses well-defined engineering concepts required to produce efficient, durable, scalable, in-budget and on-time software products.

SOFTWARE PROCESS MODEL:

- Software process can be defining as the structured set of activities that are required to develop the software system.
- To solve actual problems in an industry setting, a software engineer or a team of engineers must incorporate a development strategy that encompasses the process, methods, and tools layers.
- This strategy is often referred to as a process model or a software engineering paradigm.
- A process model for software engineering is chosen based on the nature of the project and application, the methods and tools to be used, and the controls and deliverables that are required.

Goal of Software Process Models: -

- The goal of a software process model is to provide guidance for systematically coordinating and controlling the tasks that must be performed in order to achieve the end product and their project objectives.
- A process model defines the following:
- · A set of tasks that need to be performed.
- · The inputs to and output from each task.
- · The preconditions and post-conditions for each task.
- · The sequence and flow of these tasks.

Characteristics of Software Process: -

- Software costs are concentrated in engineering (analysis and design) and not in production.
- Cost of software is not dependent on volume of production.
- Software does not wear out (in the physical sense).
- Software has no replacement (spare) parts.
- Software maintenance is a difficult problem and is very different from hardware (physical product) maintenance.
- Most software is custom-built.
- Many legal issues are involved (e.g. inter-actual property rights, liability).

Software Product: -

- A software product, user interface must be carefully designed and implemented because developers of that product and users of that product are totally different.
- In case of a program, very little documentation is expected, but a software product must be well documented.
- A program can be developed according to the programmer’s individual style of development, but a software product must be developed using the accepted software engineering principles.

Various Operational Characteristics of software are:

- **Correctness:** The software which we are making should meet all the specifications stated by the customer.
- **Usability/Learn-ability:** The amount of efforts or time required to learn how to use the software should be less. This makes the software user-friendly even for IT-illiterate people.
- **Integrity:** Just like medicines have side-effects, in the same way software may have aside-effect i.e. it may affect the working of another application. But quality software should not have side effects.

Various Operational Characteristics of software are:

- **Reliability:** The software product should not have any defects. Not only this, it shouldn't fail while execution.
- **Efficiency:** This characteristic relates to the way software uses the available resources. The software should make effective use of the storage space and execute command as per desired timing requirements.
- **Security:** With the increase in security threats nowadays, this factor is gaining importance. The software shouldn't have ill effects on data / hardware. Proper measures should be taken to keep data secure from external threats.
- **Safety:** The software should not be hazardous to the environment/life.

Difference between software process and software product: -

Software Process	Software Product
Processes are developed by individual user and it is used for personal use.	It is developed by multiple users and it is used by large number of people or Customers.
Process may be small in size and possessing limited functionality.	It consists of multiple program codes; relate documents such as SRS (Software Requirement specification), designing documents, user manuals, test cases.
Process is generally developed by process engineers.	Process is generally developed by process engineers. Therefore systematic approach of developing software product must be applied.
Software product relies on software process for its stability quality and control Only one person uses the process, hence lack of user interface	It is important than software product. Multi-user no lack of user interface.

Software Applications:

- It is somewhat difficult to develop meaningful generic categories for software applications.
- As software complexity grows, neat compartmentalization disappears.
- The following software areas indicate the breadth of potential applications:
- **System software:** System software is a collection of programs written to service other programs. Some system software (e.g., compilers, editors, and file management utilities) process complex, but determinate, information structures. Other systems applications (e.g., operating system components, drivers.)

Software Applications:

- **Real-time software:** Software that monitors/analyzes/controls real-world events as they occur is called real time.
- **Business software.** Business information processing is the largest single software application area. Discrete "systems" (e.g., payroll, accounts receivable/payable, inventory) have evolved into management information system (MIS) software that accesses one or more large databases containing business information. Applications in this area restructure existing data in a way that facilitates business operations or management decision making.

Software Applications:

- **Engineering and scientific software:** Engineering and scientific software have been characterized by "number crunching" algorithms. Applications range from astronomy to volcanology, from automotive stress analysis to space shuttle orbital dynamics, and from molecular biology to automated manufacturing.
- **Embedded software:** Embedded software resides in read-only memory and is used to control products and systems for the consumer and industrial markets. Embedded software can perform very limited functions (e.g., keypad control for a microwave oven) or provide significant function and control capability (e.g., digital functions in an automobile such as fuel control, dashboard displays, and braking systems).

Software Applications:

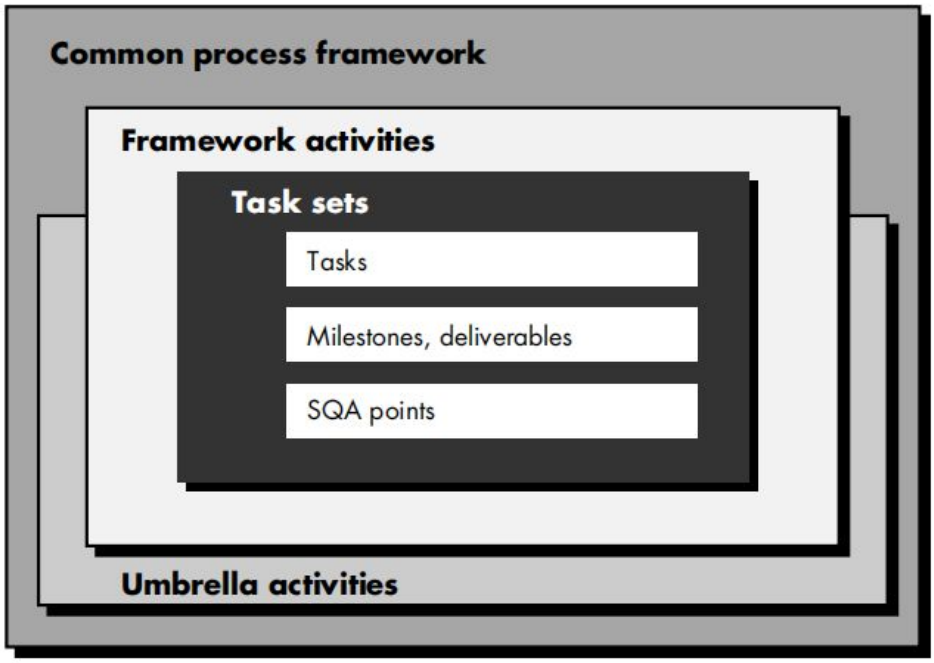
- **Web-based software:** The Web pages retrieved by a browser are software that incorporates executable instructions (e.g., CGI, HTML, Perl, or Java), and data (e.g., hypertext and a variety of visual and audio formats). In essence, the network becomes a massive computer providing an almost unlimited software resource that can be accessed by anyone with a modem.

- **Artificial intelligence software:** Artificial intelligence (AI) software makes use of nonnumerical algorithms to solve complex problems that are not amenable to computation or straightforward analysis. Expert systems, also called knowledge based systems, pattern recognition (image and voice), artificial neural networks, theorem proving, and game playing are representative of applications within this category.

Process, Methods, and Tools:

- Software engineering is a layered technology. The foundation for software engineering is the process layer.

THE SOFTWARE PROCESS:



THE SOFTWARE PROCESS:

- A software process can be characterized as shown in Figure:
- A common process framework is established by defining a small number of framework activities that are applicable to all software projects, regardless of their size or complexity.
- A number of task sets—each a collection of software engineering work tasks, project milestones,work products, and quality assurance points—enable the framework activities to be adapted to the characteristics of the software project and the requirements of the project team.

THE SOFTWARE PROCESS:

- Finally, umbrella activities—such as software quality assurance, soft ware configuration management, and measurement—overlay the process model.
- Umbrella activities are independent of any one framework activity and occur through out the process.